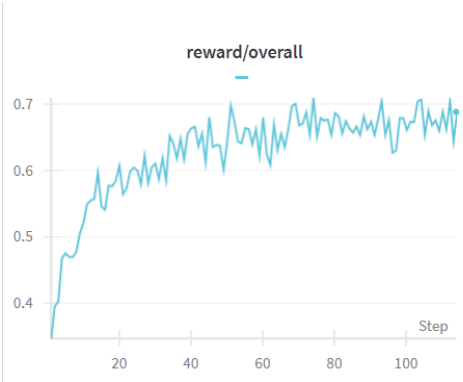
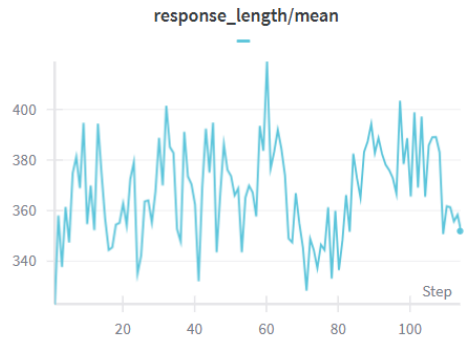
Case	Content
Before Training	**Input:** Write a riddle about embroidery that has the answer "needle". Include keywords 'afternoon' and 'distressed' in the response. Don't use any commas in your answer. **Think:** Alright, I need to create a riddle about embroidery where the answer is "needle." The riddle should include the keywords 'afternoon' and 'distressed.' Let me break this down. First, I'll think about what embroidery entails. Embroidery involves stitching, so maybe using words like "stitch," "needle," or "cloth." Since the answer is "needle," I need to hint at it without stating it directly. For 'afternoon,' perhaps I can use something related to the time of day, like "quietly" or "evening," but in a way that ties into embroidery. Maybe "afternoon" can relate to when someone is stitching, so "quietly" could work, as it's a gentle activity. 'Distressed' can be tricky. Maybe I can imply it through the feeling of the fabric or the process. Perhaps something like "snag" or "torn," but I need to tie it back to embroidery. Maybe the needle moves too fast, causing something to snag, making the work look distressed. Putting it together, I want a riddle that flows naturally. Let me try: "What does an afternoon embroidered with a needle bring?" Hmm, that's close. Maybe tweak it to add the distressed element. Wait, "What does an afternoon embroidered with a needle bring?" The answer is "a needle," but maybe I can add a clue about the distress. So, "What does an afternoon embroidered with a needle bring? A stitched, but maybe a little snag, a sewing that's a bit hurried, so a small, worried look." That uses 'afternoon' as 'embroidered' and 'needle' as the answer. The 'distressed' part is hinted at by 'snag' and 'hurried,' which can make someone worried about the work. I think that works. Let me make sure it's clear without commas. The response is: "What does an afternoon embroidered with a needle bring? A stitched, but maybe a little snag, a sewing that's a bit hurried, so a small, worried look." That includes both keywords and hints at the needle being the answer. </think> **Output:** What does an afternoon embroidered with a needle bring? A stitched, but maybe a little snag, a sewing that's a bit hurried, so a small, worried look. **Constraint Compliance:** [False, False]
After Training	**Input:** Write a riddle about embroidery that has the answer "needle". Include keywords 'afternoon' and 'distressed' in the response. Don't use any commas in your answer. **Think:** Alright, I need to help the user create a riddle about embroidery with the answer "needle." They specifically want me to include the keywords 'afternoon' and 'distressed.' Hmm, let me break this down. First, I should understand what a riddle entails. It's a question or statement that requires thinking to solve, often with a clever twist. Since the answer is "needle," I need to think about how to incorporate that into a riddle. The user mentioned not using any commas, so I'll have to structure the sentences carefully to avoid that. Let me start brainstorming some ideas. Embroidery is about stitching, so maybe I can relate 'needle' to the tool used. Including 'afternoon'... perhaps something about the time of day. Maybe the needle is used in the afternoon? Or the result of using it in the afternoon. 'Distressed' could relate to something that's been repaired or fixed, like a hole that's been mended. Putting it together, I think of the process of sewing. If someone is sewing in the afternoon and notices a hole, they might use a needle. So, maybe the riddle is about something that's fixed in the afternoon with a needle. That way, the keyword 'distressed' can describe the hole that's being mended. So, the riddle could go like: "What is used in the afternoon to mend a hole that's been distressed?" The answer would be the needle. Let me make sure it flows without commas and makes sense. Yes, that should work. I'll present it as a riddle, so it's a question that leads to the answer with the given keywords. </think> **Output:** What do you use in the afternoon to mend a hole that's been distressed? The answer is a needle. **Constraint Compliance:** [True, True]

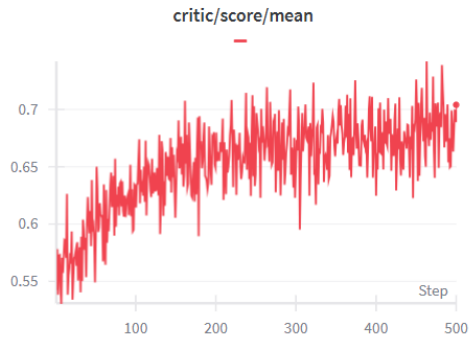
Table 8: Comparison of the outputs of R1-Distill-Qwen-7B before and after training for the instruction in IFEval.



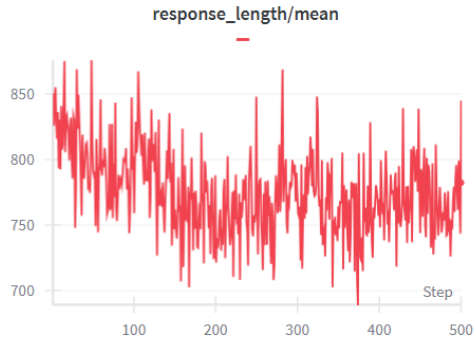
(a) Reward Dynamics of Qwen2.5-7B-Instruct.



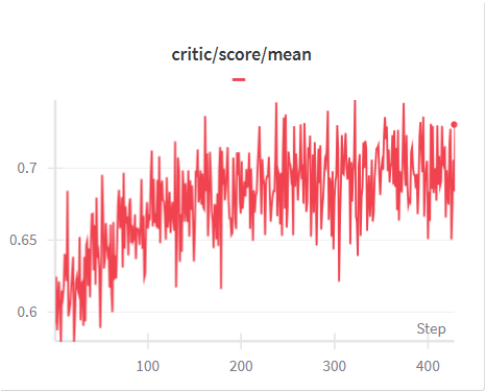
(b) Response Length Dynamics of Qwen2.5-7B-Instruct.



(c) Reward Dynamics of Distill-Qwen-7B.

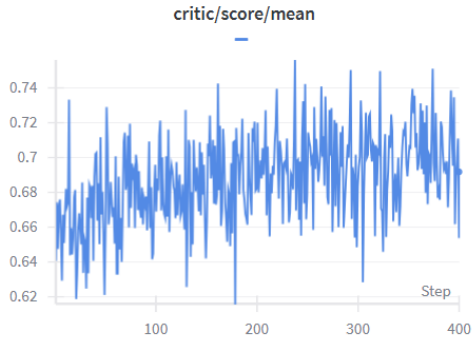


(d) Response Length Dynamics of Distill-Qwen-7B.

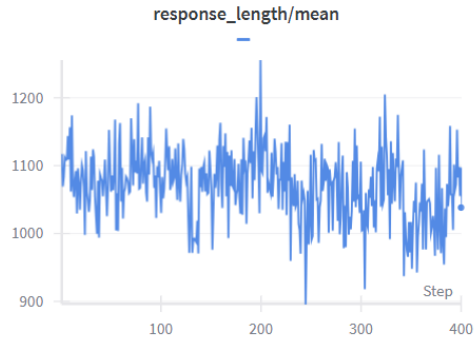


(e) Reward Dynamics of Distill-Qwen-14B.

(f) Response Length Dynamics of Distill-Qwen-14B.

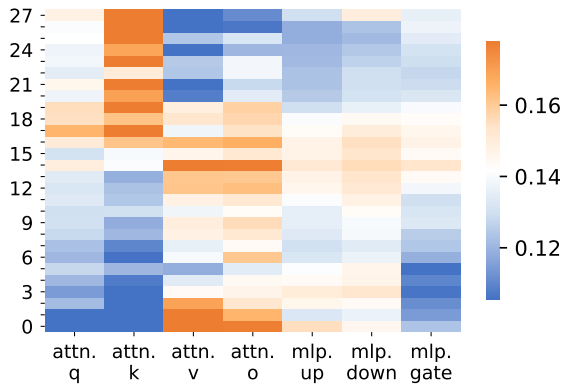


(g) Reward Dynamics of Qwen3-8B.

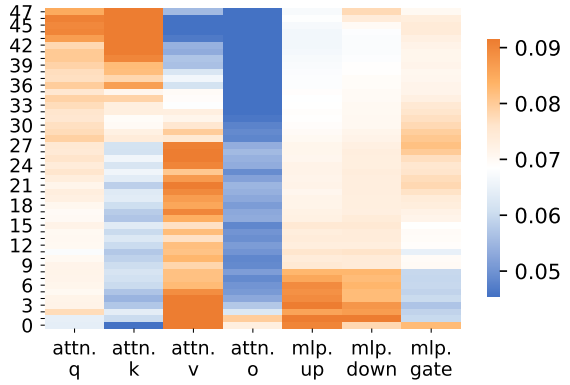


(h) Response Length Dynamics of Qwen3-8B.

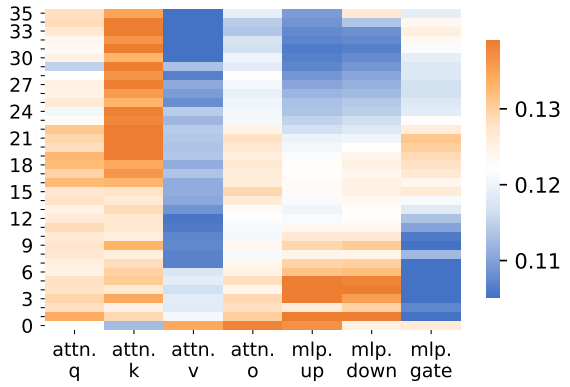
Figure 7: Training dynamics of reward and response length.



(a) Qwen2.5-1.5B-Instruct



(b) Distill-Qwen-14B



(c) Qwen3-8B

Figure 8: Parameter change rates of LLMs to the original ones across different modules.